

Part C

In this part of the test, there are two texts about different aspects of healthcare. For **questions 7-22**, choose the answer (**A, B, C or D**) which you think fits best according to the text.

Text 1: Phobia pills

An irrational fear, or phobia, can cause the heart to pound and the pulse to race. It can lead to a full-blown panic attack – and yet the sufferer is not in any real peril. All it takes is a glimpse of, for example, a spider's web for the mind and body to race into panicked overdrive. These fears are difficult to conquer, largely because, although there are no treatment guidelines specifically about phobias, the traditional way of helping the sufferer is to expose them to the fear numerous times. Through the cumulative effect of these experiences, sufferers should eventually feel an increasing sense of control over their phobia. For some people, the process is too protracted, but there may be a short cut. Drugs that work to boost learning may help someone with a phobia to 'detrain' their brain, losing the fearful associations that fuel the panic.

The brain's extraordinary ability to store new memories and forge associations is so well celebrated that its **dark side** is often disregarded. A feeling of contentment is easily evoked when we see a photo of loved ones, though the memory may sometimes be more idealised than exact. In the case of a phobia, however, a nasty experience with, say, spiders, that once triggered a panicked reaction, leads the feelings to resurge whenever the relevant cue is seen again. The current approach is exposure therapy, which uses a process called extinction learning. This involves people being gradually exposed to whatever triggers their phobia until they feel at ease with it. As the individual becomes more comfortable with each situation, the brain automatically creates a new memory – one that links the cue with reduced feelings of anxiety, rather than the sensations that mark the onset of a panic attack.

Unfortunately, while it is relatively easy to create a fear-based memory, expunging that fear is more complicated. Each exposure trial will involve a certain degree of distress in the patient, and although the process is carefully managed throughout to limit this, some psychotherapists have concluded that the treatment is unethical. Neuroscientists have been looking for new ways to speed up extinction learning **for that same reason**.

One such avenue is the use of 'cognitive enhancers' such as a drug called D-cycloserine or DCS. DCS slots into part of the brain's 'NMDA receptor' and seems to modulate the neurons' ability to adjust their signalling in response to events. This tuning of a neuron's firing is thought to be one of the key ways the brain stores memories, and, at very low doses, DCS appears to boost that process, improving our ability to learn. In 2004, a team from Emory University in Atlanta, USA, tested whether DCS could also help people with phobias. A pilot trial was conducted on 28 people undergoing specific exposure therapy for acrophobia – a fear of heights. Results showed that those given a small amount of DCS alongside their regular therapy were able to reduce their phobia to a greater extent than those given a placebo. Since then, other groups have replicated the finding in further trials.

For people undergoing exposure therapy, achieving just one of the steps on the long journey to overcoming their fears requires considerable perseverance, says Cristian Sirbu, a behavioural scientist and psychologist. Thanks to improvement being so slow, patients – often already anxious – tend to feel they have failed. But Sirbu thinks that DCS may make it possible to tackle the problem in a single 3-hour session, which is enough for the patient to make real headway and to leave with a feeling of satisfaction. However, some people have misgivings about this approach, claiming that as it doesn't directly undo the fearful response which is deep-seated in the memory, there is a very real risk of relapse.

Rather than simply attempting to overlay the fearful associations with new ones, Merel Kindt at the University of Amsterdam is instead trying to alter the associations at source. Kindt's studies into anxiety disorders are based on the idea that memories are not only vulnerable to alteration when they're first laid down, but, of key importance, also at later retrieval. This allows for memories to be 'updated', and these amended memories are re-consolidated by the effect of proteins which alter synaptic responses, thereby maintaining the strength of feeling associated with the original memory. Kindt's team has produced encouraging results with arachnophobic patients by giving them propranolol, a well-known and well-tolerated beta-blocker drug, while they looked at spiders. This blocked the effects of norepinephrine in the brain, disrupting the way the memory was put back into storage after being retrieved, as part of the process of reconsolidation. Participants reported that while they still don't like spiders, they were able to approach them. Kindt reports that the benefit was still there three months after the test ended.

7. In the first paragraph, the writer says that conventional management of phobias can be problematic because of
- (A) the lasting psychological effects of the treatment.
 - (B) the time required to identify the cause of the phobia.
 - (C) the limited choice of therapies available to professionals.
 - (D) the need for the phobia to be confronted repeatedly over time.
8. In the second paragraph, the writer uses the phrase 'dark side' to reinforce the idea that
- (A) memories of agreeable events tend to be inaccurate.
 - (B) positive memories can be negatively distorted over time.
 - (C) unhappy memories are often more detailed than happy ones.
 - (D) unpleasant memories are aroused in response to certain prompts.
9. In the second paragraph, extinction learning is explained as a process which
- (A) makes use of an innate function of the brain.
 - (B) encourages patients to analyse their particular fears.
 - (C) shows patients how to react when having a panic attack.
 - (D) focuses on a previously little-understood part of the brain.
10. What does the phrase 'for that same reason' refer to?
- (A) the anxiety that patients feel during therapy
 - (B) complaints from patients who feel unsupported
 - (C) the conflicting ethical concerns of neuroscientists
 - (D) psychotherapists who take on unsuitable patients

11. In the fourth paragraph, we learn that the drug called DCS
- (A) is unsafe to use except in small quantities.
 - (B) helps to control only certain types of phobias.
 - (C) affects how neurons in the brain react to stimuli.
 - (D) increases the emotional impact of certain events.
12. In the fifth paragraph, some critics believe that one drawback of using DCS is that
- (A) its benefits are likely to be of limited duration.
 - (B) it is only helpful for certain types of personality.
 - (C) few patients are likely to complete the course of treatment.
 - (D) patients feel discouraged by their apparent lack of progress.
13. In the final paragraph, we learn that Kindt's studies into anxiety disorders focused on how
- (A) proteins can affect memory retrieval.
 - (B) memories are superimposed on each other.
 - (C) negative memories can be reduced in frequency.
 - (D) the emotional force of a memory is naturally retained.
14. The writer suggests that propranolol may
- (A) not offer a permanent solution for patients' phobias.
 - (B) increase patients' tolerance of key triggers.
 - (C) produce some beneficial side-effects.
 - (D) be inappropriate for certain phobias.

Text 2: Challenging medical thinking on placebos

Dr Damien Finniss, Associate Professor at Sydney University's Pain Management and Research Institute, was previously a physiotherapist. He regularly treated football players during training sessions using therapeutic ultrasound. 'One particular session', Finniss explains, 'I treated five or six athletes. I'd treat them for five or ten minutes and they'd say, "I feel much better" and run back onto the field. But at the end of the session, I realised the ultrasound wasn't on.' It was a light bulb moment that set Finniss on the path to becoming a leading researcher on the placebo effect.

Used to treat depression, psoriasis and Parkinson's, to name but a few, placebos have an image problem among medics. For years, the thinking has been that a placebo is useless unless the doctor convinces the patient that it's a genuine treatment – problematic for a profession that promotes informed consent. However, a new study casts doubt on this assumption and, along with a swathe of research showing some remarkable results with placebos, raises questions about whether they should now enter the mainstream as legitimate prescription items. The study examined five trials in which participants were told they were getting a placebo, and the conclusion was that doing so honestly can work.

'If the evidence is there, I don't see the harm in openly administering a placebo,' says Ben Colagiuri, a researcher at the University of Sydney. Colagiuri recently published a meta-analysis of thirteen studies which concluded that placebo sleeping pills, whose genuine counterparts **notch up** nearly three million prescriptions in Australia annually, significantly improve sleep quality. The use of placebos could therefore reduce medical costs and the burden of disease in terms of adverse reactions.

But the placebo effect isn't just about fake treatments. It's about raising patients' expectations of a positive result; something which also occurs with real drugs. Finniss cites the 'open-hidden' effect, whereby an analgesic can be twice as effective if the patient knows they're getting it, compared to receiving it unknowingly. 'Treatment is always part medical and part ritual,' says Finniss. This includes the austere consulting room and even the doctor's clothing. But behind the performance of healing is some strong science. Simply believing an analgesic will work activates the same brain regions as the genuine drug. 'Part of the outcome of what we do is the way we interact with patients,' says Finniss.

That interaction is also the focus of Colagiuri's research. He's looking into the 'nocebo' effect, when a patient's pessimism about a treatment becomes self-fulfilling. 'If you give a placebo, and warn only 50% of the patients about side effects, those you warn report more side effects,' says Colagiuri. He's aiming to reverse that by exploiting the psychology of food packaging. Products are labelled '98% fat-free' rather than '2% fat' because positive reference to the word 'fat' puts consumers off. Colagiuri is deploying similar tactics. A drug with a 30% chance of causing a side effect can be reframed as having a 70% chance of not causing it. 'You're giving the same information, but framing it a way that minimises negative expectations,' says Colagiuri.

There is also a body of research showing that a placebo can produce a genuine biological response that could affect the disease process itself. It can be traced back to a study from the 1970s, when psychologist Robert Ader was trying to condition taste-aversion in rats. He gave them a saccharine drink whilst simultaneously injecting Cytoxan, an immune-suppressant which causes nausea. The rats learned to hate the drink due to the nausea. But as Ader continued giving it to them, without Cytoxan, they began to die from infection. Their immune system had 'learned' to fail by repeated pairing of the drink with Cytoxan. Professor Andrea Evers of Leiden University is running a study that capitalises on this conditioning effect and may benefit patients with rheumatoid arthritis, which causes the immune system to attack the joints. Evers' patients are given the immunosuppressant methotrexate, but instead of always receiving the same dose, they get a higher dose followed by a lower one. The theory is that the higher dose will cause the body to link the medication with a damped-down immune system. The lower dose will then work because the body has 'learned' to curb immunity as a placebo response to taking the drug. Evers hopes it will mean effective drug regimes that use lower doses with fewer side effects.

The medical profession, however, remains less than enthusiastic about placebos. 'I'm one of two researchers in the country who speak on placebos, and I've been invited to lecture at just one university,' says Finniss. According to Charlotte Blease, a philosopher of science, this antipathy may go to the core of what it means to be a doctor. 'Medical education is largely about biomedical facts. 'Softer' sciences, such as psychology, get marginalised because it's the hard stuff that's associated with what it means to be a doctor.' The result, says Blease, is a large, placebo-shaped hole in the medical curriculum. 'There's a great deal of medical illiteracy about the placebo effect ... it's the science behind the art of medicine. Doctors need training in that.'

15. A football training session sparked Dr Finniss' interest in the placebo effect because
- (A) he saw for himself how it could work in practice.
 - (B) he took the opportunity to try out a theory about it.
 - (C) he made a discovery about how it works with groups.
 - (D) he realised he was more interested in research than treatment.
16. The writer suggests that doctors should be more willing to prescribe placebos now because
- (A) research indicates that they are effective even without deceit.
 - (B) recent studies are more reliable than those conducted in the past.
 - (C) they have been accepted as a treatment by many in the profession.
 - (D) they have been shown to relieve symptoms in a wide range of conditions.
17. What is suggested about sleeping pills by the use of the verb 'notch up'?
- (A) they may have negative results
 - (B) they could easily be replaced
 - (C) they are extremely effective
 - (D) they are very widely used
18. What point does the writer make in the fourth paragraph?
- (A) The way a treatment is presented is significant even if it is a placebo.
 - (B) The method by which a drug is administered is more important than its content.
 - (C) The theatrical side of medicine should not be allowed to detract from the science.
 - (D) The outcome of a placebo treatment is affected by whether the doctor believes in it.

19. In researching side effects, Colagiuri aims to
- (A) discover whether placebos can cause them.
 - (B) reduce the number of people who experience them.
 - (C) make information about them more accessible to patients.
 - (D) investigate whether pessimistic patients are more likely to suffer from them.
20. What does the word 'it' in the sixth paragraph refer to?
- (A) a placebo treatment
 - (B) the disease process itself
 - (C) a growing body of research
 - (D) a genuine biological response
21. What does the writer tell us about Ader's and Evers' studies?
- (A) Both involve gradually reducing the dosage of a drug.
 - (B) Evers is exploiting a response which Ader discovered by chance.
 - (C) Both examine the side effects caused by immunosuppressant drugs.
 - (D) Evers is investigating whether the human immune system reacts to placebos as Ader's rats did.
22. According to Charlotte Blease, placebos are omitted from medical training because
- (A) there are so many practical subjects which need to be covered.
 - (B) those who train doctors do not believe that they work.
 - (C) they can be administered without specialist training.
 - (D) their effect is more psychological than physical.